Predicting Below Knee Muscle Forces from Ground Reaction Forces in Older and Younger Populations Using Sequence- and Attention-Based Models

Foot & Ankle Mechanics and Morphology Lab

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Introduction

Mobility decline is an increasingly prevalent problem. It significantly impacts the quality of life for millions and places a multi-billion dollar burden on the healthcare system [1,2].

Abnormal joint loading drives osteoarthritis progression [3].
Weak ankle plantarflexors are associated with poor balance and increased fall risk [4].

Recording these fall-risk indicating signals is nearly impossible in living people. The best way to estimate them requires a full motion capture laboratory and musculoskeltal simulation [3].

Objective: Create a deep learning model that uses force plate data to predict muscle forces crossing the ankle and joint contact forces at the ankle and knee.

What if a single force plate with a piece of software could quickly predict these quantities in a clinical or rehabilitation setting?

Methods

Dataset	Uhlrich [7]	Silder Old Adult [6]	Silder Young Adult [6]
Subjects	10	21	21
Trial Duration	~6 minutes	~15 seconds	~15 seconds
Number of gait cycles	13022	554	502
Measurement Device	Instrumented treadmill	3 discrete force plates	3 discrete force plates
Ground truth force estimation	EMG-informed static optimization	Static optimization (no EMG data)	Static optimization (no EMG data)
Demographics	4 F · 26 ± 4 yrs	13 F · 72.5 ± 5.1 yrs	12 F · 26 ± 3.1 yrs

Purpose: Uhlrich data was used to validate this approach against gold-standard targets.

Purpose: Silder data extended the approach to older and younger adult populations with standard labels.

Data Processing

Stance-phase segmentation
Body mass normalization
Subject-stratified splits
train/test/val~80/10/10

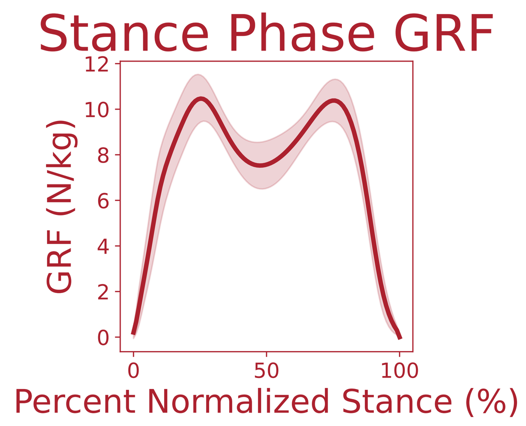


Figure 1: Compiled stance phase segments of vertical ground reaction forces. The dark red line is the mean of the normalized vertical ground reaction forces, and the shading indicates ± one standard deviation. There were 1056 individual segments in this dataset. Ground reaction forces are normalized to body weight.

Inputs

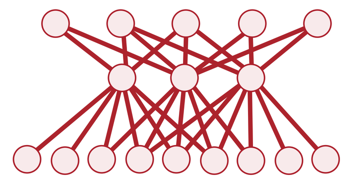
Force Plate Quantities (measured):
Ground reaction forces (GRF) — x / y / z
Center of pressure — x / z

Training only

Ground-truth labels
(Calculated via static optimization)
Muscle forces
Joint contact forces

Models

Long Short-Term Memory (LSTM)
LSTM + Attention
Convolutional Neural Net (CNN)-LSTM
Transformer



Outputs

Predicted Quantities:
Muscle forces
Joint contact forces

Results

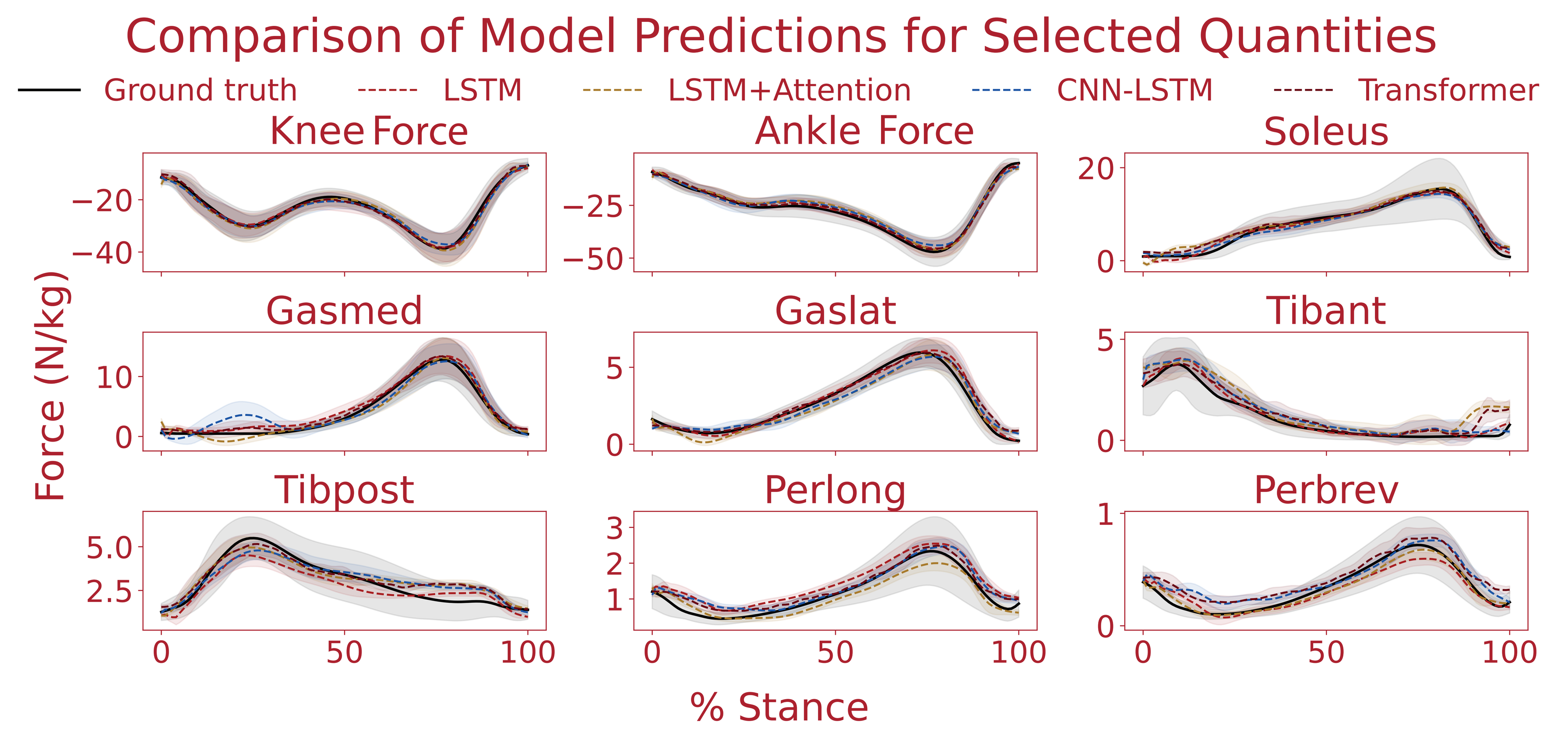
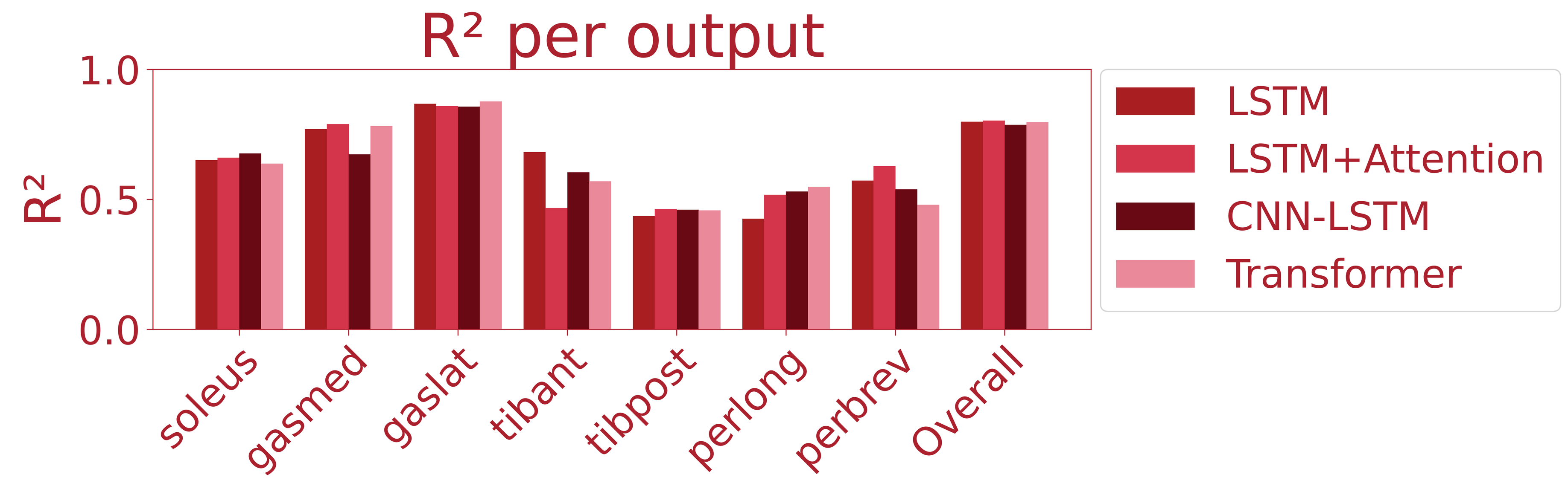


Figure 2: Predicted versus ground-truth force across stance for seven ankle-crossing muscles (Soleus: soleus, Gasmed: medial gastrocnemius, Gaslat: lateral gastrocnemius, Tibant: tibialis anterior, Tibpost: tibialis posterior, Perlong: peroneus longus, Perbrev: peroneous brevis) .and the vertical components of ankle (Ankle Force) and knee (Knee Force) joint contact force (normalized to body mass, N/kg). Black: ground truth from static optimization; colored: model predictions from ground reaction forces alone; shaded bands: ±1 SD across test gait cycles. Predictions reproduce waveform magnitude and timing, particularly for the well-constrained triceps surae (soleus, medial and lateral gastrocnemius) and both joint contact forces.

Takeaway: These models capture non-linear biomechanical relationships well enough to estimate difficult-to-measure joint forces and major muscle forces from a single force plate, with clinically relevant accuracy.

Figure 3: Coefficient of determination (R^2) for muscle-force predictions, by muscle and model. The triceps surae and tibialis anterior are predicted with high R^2; the deep redundant synergists (tibialis posterior and the peroneals) show lower R^2, consistent with the non-uniqueness of their static-optimization labels rather than a model limitation. The CNN-LSTM performed best overall with this muscle subset.



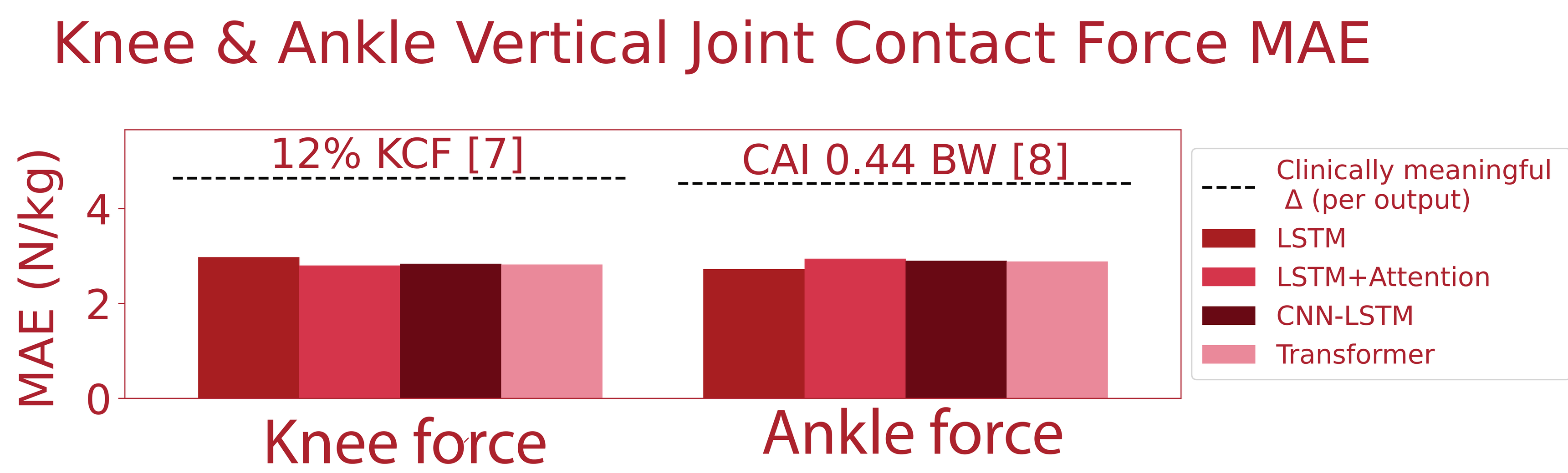


Figure 4: Mean absolute error of predicted vertical joint contact force at the ankle and knee, by model. Dashed lines mark documented clinically meaningful differences per joint: at the knee, the 12% late-stance reduction in knee contact force (KCF) associated with reduced cartilage loss [3, 7]; at the ankle, the 0.44 BW (≈4.3 N/kg) lower compressive peak in chronic ankle instability (CAI), the principal pathway to post-traumatic ankle osteoarthritis [8]. Prediction error from force plate quantities alone falls well below both thresholds. Lines are peak-magnitude references; bars are whole-cycle MAE.

Discussion

- Models predicted joint contact forces and muscle forces from force-plate data alone, with error below documented clinically meaningful differences (Fig. 4).
- This provides access to these difficult-to-measure forces in clincal and rehab settings using only the data from a single force plate.
- Triceps surae and joint contact forces predicted accurately; redundant synergists (tibialis posterior, peroneals) less so, reflects how well-constrained each output is.
- Static optimization cannot uniquely partition force among those synergists, so their labels are inherently uncertain. The lower accuracy reflects label ambiguity, not model failure, and reinforces joint contact force as the more robust target.

Limitations

Ground-truth forces were estimated via static optimization, not measured; in vivo contact force requires instrumented implants.
Older and younger groups did not differ significantly in ground-truth peak force in this cohort. Therefore, models are accurate estimators, not detectors of age-related change.

Future Work

Validate against instrumented-implant data to anchor ground truth force estimates to the true gold standard.
Apply to clinical populations with established loading differences to test whether this precision detects differences that matter.

References & Acknowledgments

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